AI-driven Policy Simulation & Experimentation: a review

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Abstract.

This paper reviews the impact of Artificial Intelligence (AI) on the simulation and outcomes of policies, with a specific focus on Agentic Ai and multi-agent models. AI-driven policy simulation uses AI to model, test, and predict the outcomes of national government policies before implementation, which reduces risks and enhances evidence-based decision-making processes. AI can be used effectively to simulate complex scenarios and optimize resource allocation by creating 'digital twins' of socio-political and economic systems.

Major Policy Developments – Traditional Policy:

- Policy developments have involved the use of advice from human experts by governments and think tanks. Policies are then deployed, and the outcomes are evaluated.
- AI-driven policy creates 'digital twins' which are data-rich models of real-world systems with simulations that are used to test the potential impact of new policies and legal frameworks before their full implementation and deployment.

AI has become increasingly integrated into socio-political and economic processes during the 2020s. Hence, modern policymakers have pivoted towards greater reliance on AI-driven policy simulations.

Firstly, the paper's contributions are to present a taxonomy of AI's impact on policy development. Secondly, to review AI policy simulation and to present a cast study of an AI-driven policy simulation platform. Thirdly, this paper examines the impact of Ai automation on employment. In addition, this paper analyzes the emergence of modern technological developments that have had a profound impact on contemporary national governments and policymakers.

1. Introduction

Modern society is on the threshold of the 4th Industrial (AI) Revolution, which is already having significant impacts on socio-political and economic issues, as well as on national government policies (Treleaven and Brown, 2025). In the 21st Century, two distinct approaches to the creation and refinement of public policies have therefore emerged: Traditional policy development and AI-driven policy simulation (Lee & Dai, 2025). The former is centered on the long-established reliance of national governments, think tanks, and organizations on experts with specialized knowledge of socio-political and economic events and issues (Dexter, 2006; Von Soest, 2022).

Expert advice is supported by testing and analysis of policies, which occur through economic modelling, scenario planning, and data analysis using static tools such as regression models (Von Capano et al., 2022). The traditional development of policies has also involved the formulation of a largely linear policy cycle. This is a framework in which problems are identified as agendas and strategies are set before the deployment of actionable decision-making processes by multiple policymakers (Capano & Pritoni, 2022).

The main focus of this traditional approach to policy, therefore, lies in incorporating the cultural, political, and institutional nuances of human experts, which are typically difficult for machine learning models to fully address (Roche, Wall, and Lewis, 2022). However, there is an academic consensus that this approach has a significant weakness in addressing policy issues, as human experts are inherently biased and often fail to incorporate emergent behaviors in large populations (Kynn, 2007; Fischer et al., 2014; Karimi et al., 2025).

Hence, this paper examines the emergence of the modern AI-driven policy approach in the 2020s. This approach has been driven by machine learning, particularly multi-agent modelling, powered by large language models (LLMs). Consequently, the policy simulation research of this paper has involved AI machine learning, which is specifically focused on using technology to innovatively create 'digital twins' and experimental environments that can mimic real-world dynamics.

Digital twins are computer-based models of specific entities and policies (Mihai et al., 2022). This includes physical objects, processes, or systems in which AI agents interact autonomously and represent a wide range of stakeholders from governments to citizens (Kreuzer et al., 2024). The systematic use of digital twins enables policymakers to test policies by simulating countless scenarios and parameters and

observing emergent outcomes, such as socio-political and economic behavior (Marucci et al., 2020). As the innovative use of AI drives this emerging (policy) approach, common techniques include LLM-based Agentic AI, multi-agent systems, and self-improving machine learning models.

The authors of this paper acknowledge that AI-driven policies have alimitations that must be comprehensively addressed to ensure long-term success. One such limitation is that AI is often prone to hallucinations and susceptible to bias in training data and model design (Salvagno, Taccone, and Gerli, 2023). However, maintaining humans in the loop is key to ensuring the success of the AI-driven policy approach (Mosquiera-Ray, 2022).

The main strength of this approach is that AI is often less expensive than human experts and can better manage complex policy issues and outcomes (Nordstrom, 2021). A further strength of this approach is that, because AI is less reliant on human biases, it can analyze data more effectively (Chuong et al., 2024). It is therefore evident that AI has become inextricably linked to policymaking, and its increased implementation and use will become essential to determining the simulations and outcomes of modern policies.

In summary, the so-called “human in the loop” approach allows AI-driven policy simulation to provide policy professionals with a valuable experimental tool combining the strengths of traditional insight and AI-driven experimentation (see Table 1).

Next, as background, we review the technologies underpinning AI-driven policy.

AI & Emerging Technology.

As discussed, AI is becoming a major factor in modern socio-political and economic issues, as well as in national government policies. Hence, multiple emerging technologies are driving innovation and transformative change on a global scale in the 2020s.

Machine Learning.

Machine learning (ML) has rapidly emerged as a powerful tool, using data analysis to automate model building and enable computers to make independent decisions (Sharifani and Amini, 2023). In this section, we categorize and elucidate the subfields of ML as follows: Classic ML, Generative AI, Agentic AI, and LLMs.

ML techniques can be divided into:

- **Classic ML:** classic ML techniques primarily involve statistical and connectionist models that rely on feature engineering and discriminative learning. These methods typically require carefully curated datasets and substantial domain knowledge to extract relevant features (Faouzi and Colliot, 2023).
- **Generative AI (GenAI):** GenAI focuses on creating systems capable of dynamically producing text, images, audio, video, code, or other modalities in a manner that appears strikingly human like. LLMs are essentially a subset of GenAI for human-like text and speech:
- *LLM* used to generate text or complete tasks based on their training data and pre-defined capabilities of LLMs (e.g. Traditional LLMs). In addition, *LLM Agents* introduce features to LLMs like planning, memory, and tool usage to achieve greater autonomy and handle more complex, multi-step tasks.
- *Agentic AI* - autonomous AI systems that use LLMs and other components to perceive, reason, plan, and act independently toward complex goals with minimal human intervention, acting more like proactive digital team

The relationship between Agentic AI and self-improving models is deeply intertwined and increasingly central to ML applications. Self-improving models refer to AI systems (often large language models) that can autonomously enhance their own performance—through techniques like self-improvement, code rewriting, even meta-learning—without requiring retraining. Agentic AI, in contrast, describes proactive, autonomous systems that pursue goals independently.

Unpacking the key underlying GenAI and self-improving models:

- **Artificial Neural Networks (ANNs):** ANNs are a type of machine learning (ML) process that uses interconnected nodes or neurons in a layered structure “inspired” by the human brain [McCulloch & Pitts, 1943]. An ANN uses training examples and learns by adjusting a “weight” associated with each connection linking two neurons (Rumelhart et al, 1986).

- **Deep Learning:** uses ANNs with multiple layers (i.e., deep neural networks) to analyze data and learn complex patterns (LeCun et al, 2015).
- **Transformers:** a type of deep learning ANN model that is used for NLP tasks and can learn long-range dependencies between words in a sentence (Vaswani et al, 2017; Devlin et al, 2019; Lin et al, 2021).
- **Large Language Models (LLM)**: uses deep learning techniques and massive datasets to understand, summarize, generate, predict new content, and hold a dialog (Sharkey & Treleaven, 2025).
- **Self-improving AI models**: are AI systems capable of enhancing their own algorithms of code, architecture, or training data autonomously, with little to no human intervention. Examples include:
- *Agentic AI* - take independent action to complete tasks with little to no human oversight.

Multi-Agent models - here multiple, specialized AI entities (agents) work together to solve complex problems.

Typically, classic ML requires direct engagement with a wide range of datasets and substantial domain knowledge to extract relevant features, as well as scraping data, model building, and extensive testing (Tufail et al., 2023).

In the 2020s, Generative AI (GenAI) became globally recognized as it is foundational to popular generative chatbots such as ChatGPT (Chiu, 2023). The recent success of GenAI has been facilitated by its systems that are dynamically capable of producing a wide range of modalities (e.g., text, imagery, audio, and video), which coherently engage with individuals and generate human-like speech and texts (Zeleke et al., 2025).

LLMs are inextricably linked to GenAI. These models leverage deep learning techniques, using their training data and capabilities to generate text, complete tasks, and ensure that complex language data is fully understood (Hadi et al., 2024). In addition, the use of LLM agents represents a technological advancement in LLMs by introducing features such as planning, memory, and tool usage (Yang et al., 2025). Hence, there is an academic consensus that equipping LLMs with the ability to handle more intricate tasks and engage in decision-making processes has enabled these computational models to attain greater autonomy rapidly (Handler, 2023; Zheng et al., 2025; Khan et al., 2026).

Agentic AI is an autonomous system that requires minimal human intervention and uses LLMs to proactively perceive, reason, plan, and act towards complex goals (Hosseini and Seilani, 2025). In addition, Agentic AI is inextricably linked to self-improving models that can autonomously enhance their performance through practices such as code rewriting and meta-learning (Bar, 2025).

Self-Improving Models:

Self-improving AL models show that ML developments form a progression toward more autonomous, adaptive, and powerful intelligence. Self-improving models—also called self-evolving, self-adapting, or recursive self-improvement (RSI) systems autonomously enhance their own capabilities by generating their own training data, critiquing, and refining outputs, modifying code or hyperparameters, or running internal optimization loops without constant external supervision.

The relationship between self-improving models, agentic ai and multi-agent models form a tightly coupled progression. Broadly, agentic AI supplies the autonomy and execution engine needed for self-improvement. Multi-agent architectures speed the process by enabling division of labor, collective intelligence, and evolutionary exploration. Self-improving models turn agentic systems and multi-agent processes into lifelong learners. In essence: Agentic AI is the substrate; multi-agent processes are the accelerator; and self-improvement is the capability.

The decentralized approach of Distributed Ledger Technology (DLT) effectively supports secure collaboration and cross-organizational workflows (Anthony Junior, 2021). While it is possible to implement policy simulation tools without using DLT, there is an academic consensus that tokenization, smart contracts, and DLT all provide comprehensive security and greater automation (Soltani et al., 2022; Vionis and Kotsilieris, 2023; Cali et al., 2025).

In Figure 1, we illustrate the key components of a distributed ledger.

The timestamp is the time and date of the block's creation. Secondly, the hash is a unique digital identifier for transactions, used to ensure the integrity of individual blocks. Lastly, nonces are values used only once

and are often employed to enhance security by ensuring that cryptographic hash algorithms and authentication protocols are protected (Mostafa, Lee, and Peker, 2020).

FEDERATED COLLABORATION:

There has been a recent rise in multi-dataset integration and multi-partner and multi-dataset collaboration, as well as in the governance of policy design (Ruijer, 2021). These processes require the implementation of federated computing and data infrastructure

Broadly speaking there are three technology architecture types to support multi-dataset and multi-partner collaboration:

- **Shared database platforms**: a "data lake" centralized repositories that store raw data in its original format, to enable broad data exploration. Privacy and regulation preclude its use for sensitive health data (Ref).
- **Trusted Research Environments (TRE)**: a secure, controlled digital environment where researchers can access and analyze a sensitive dataset. Currently used in healthcare but may preclude integrating multiple datasets (O'Donovan et al., 2023).
- **Federated Computing (FC)**: supports multi-partner, multidiscipline and multi-datasets analytics. In simple terms, FC is an "orchestrator" infrastructure for connection of TREs and analytics, facilitating uniform access of data and algorithms (Fenoglio & Treleaven, 2026).

Federated computing has four foundational pillars for sovereignty-preserving integration:

- **Distributed data assets:** local control: Ensures that data remains under the jurisdiction and control of its original owner, avoiding centralization and enabling compliance with privacy and regulatory constraints. (e.g., data mesh).
- **Federated services:** in-place computation: Brings algorithms to the data rather than moving data to centralized locations, allowing useful computation to occur locally without exposing sensitive information (e.g., federated learning frameworks).
- **Standardized APIs:** interoperability: Defines how systems communicate across organizational boundaries using secure, modular APIs that support interoperability and composability across heterogeneous infrastructures (e.g., OpenAPI-defined endpoints).
- **Decentralized governance:** multi-party control and dynamic governance: Enables peer systems to collaborate without a central authority, using transparent control mechanisms that respect

autonomy and support jointly enforced policy constraints (e.g., blockchain or other distributed-consensus substrates).

3.AI-Driven Policy Simulation Technology.

In this section, we review AI-driven policy simulation technology and modelling techniques. Contemporary policy simulation technology is therefore categorized as follows: Modelling paradigm, integration level (AI depth), scale/purpose, and user/ownership.

Core Modelling ParadigmsTHE CORE MODELLING PARADIGM:

Traditionally, tThe computer modelling paradigm for systems, traditionally, has been the Agent-based paradigm (Ref). However, Agents have been superseded by the rise of Agentic AI and self-improving models recently:

- **Agent-Based Models (ABM)**: models the actions and interactions of autonomous "agents" (individuals, groups, or organizations) to understand their emergent system-level behavior. Applications include social/economic/policy foresight.
- **System Dynamics Models (SDM)**: visual representations used to understand, simulate, and analyze how complex systems behave. Applications include causal understanding and broad forces.
- **Agentic AI:** an autonomous LLMs agent capable of perceiving environments, planning, making decisions, and executing complex, multi-step workflows to achieve goals with minimal human oversight.

Multi-Agent Systems (MAS): multiple intelligent, autonomous agents that interact with each other within a shared environment to solve complex problems, often acting like a coordinated team. These agents employ GenAI, LLMs, multi-agent orchestration, and self-improvement. Applications include reasoning and planning.

As discussed, for policy simulation these technologies are used to support:

Digital Twins: a virtual model digitally replicating an object, a system, or a process. Real-time, data-driven virtual replicas of systems/cities/populations. Often hybrid with ABM/SDM + AI for predictive "living" simulations.

Overall, these technologies are used to support policy simulation. Furthermore, these technological advancements support digital twins, which are virtual models that replicate an object, system, or process. The models of systems, cities, and populations are data-driven, operate in real time, and are hybrid, combining ABM/SDM with AI for predictive 'living' simulations.

Table 2 from Grok presents a simple taxonomy of policy simulation technology.

In this section, the paper examines the creation and application of AI-driven policy simulation platforms, which are used as 'sandboxes.' This section also examines the creation of a digital twin testing environment designed to enable the safe testing and modelling of conceptual policies and their potential socio-political and economic impacts before full implementation.

The tools for AI-driven policy simulation range from simple interactive web-based simulators to a digital twin testing environment that uses advanced Agentic AI and self-improving models. Hence, the policy platforms of this research project are designed to fully support scenario analysis, outcome forecasting (e.g., financial costs and environmental effects), and participatory decision-making processes for stakeholders, from national governments to policymakers.

Policy Simulation Application Areas.

In general, policy simulation platforms are used in specialist areas to reduce the costs and risks of in-person policy trials and to improve the quality of outcomes intended for national governments to implement.

- **Public Health & Social Policy:** Policy simulations are designed to cover responses to pandemics and epidemics. They can also address resource allocation and public engagement in healthcare systems, as well as the adoption of innovative technology such as genetic engineering.

- **Economic & Fiscal Policy:** Policy simulations can address the effects of taxation, labour market reforms, regional transfers, and inequality, such as food insecurity rates across countries.

- **Climate & Environmental Policy:** AI-driven tools can fully investigate and assess the impact of policies, such as modelling emissions trading, energy transitions, and adaptation, and strategically allocate investment resources.

- **Urban & Infrastructure Planning:** Policy simulations can address concerns regarding traffic optimization, disaster response, green infrastructure, and reconstruction forecasting.

- **Energy Markets & Resource Management:** Because national energy policies are a significant geopolitical issue, policy simulation can address market dynamics, resource scarcity, and energy infrastructure.

Technology & AI Governance: Policy simulations can address concerns about new regulations and legislation for emerging technologies. They can also simulate the socio-political impact of AI and the effectiveness of regulatory frameworks such as the 2024 AI Act in the European Union.

- **Public Policy Experimentation:** This enables policymakers and experts to oversee the development of digital twin simulators and conduct full policy experiments before deployment.
- **Participatory Policy Engagement:** This is facilitated by 'Polco' simulation tools that enable citizens and civil servants to gain insights into policies and decision-making processes to enable positive benefits for their communities.
- **Specialist Domain Simulators:** These models complex issues in key systems (e.g., healthcare, and urban planning) to help policymakers predict the outcomes of policy decisions.

Public Policy Experimentation

Public policy experimentation is a structured and evidence-based approach. It enables policymakers to test, refine, and evaluate the frameworks of new policies and their outcomes. Furthermore, policy experimentation platforms aim to reduce uncertainty, minimize risks, and identify key issues by using methodologies (e.g., trials) to measure the scope of policy impact and foster innovative approaches. Several examples of public policy experimentation platforms are therefore elucidated as follows:

- **Generative Agent-Based Framework for Policy Simulation & Evaluation (GPLab):** This this framework is a general computational platform that uses GenAI (LLMs) and Agent-Based Modelling (ABM) to simulate, test, and evaluate public policies.

- **Forio Epicenter**: This platform enables policymakers to build custom models and scenarios, analyze combinatorial effects, select initiatives, compare scenarios, and share results. Organizations like the World Bank, CDC, and ReThink Health often use it.

- **Interactive Social Policy Simulator (ISPS):** These are platforms built using analytics for key tasks such as visualizing administrative data.

- **AnyLogic:** This is a social process simulation platform that uses multi-agent models to evaluate social processes, demographics, urbanization, and healthcare policies. It is also designed to support large-scale synthetic populations and scenario testing effectively.

- **Politika:** is a web-based meta-simulation developed by Policy Cloud. It relies on social simulation to engage with policy analysis and design, as well as simulations, to successfully achieve its objectives.

- **EvalCommunity:** An AI-powered policy simulator used to design and test evidence-based policies, generating policy briefs and recommendations. It enables users to define their own objectives, select data, configure parameters, and manage AI simulations to determine the scope of costs, impacts, and risks involved in policy outcomes. Furthermore, EvalCommunity uses census data, satellite imagery, and administrative records to model concrete outcomes for objectives such as facilitating substantial economic growth.

- **gentTorch:** is a (MIT Media Lab) is a platform that scales LLM-guided agent simulations for population-level policy testing.

Other simulation platforms include Governing with Foresight (Courtney Fallow), Policy Priority Inference (PPI), Aaru Simulation, ESR's Digital Twin Platform (New Zealand), Real-Time Policy Simulator (various implementations), Generative Scenario Simulation for Policy Impact, AI Economist/Salesforce-style frameworks, and Monetary Policy Simulator.

Participatory Policy Engagement

Polco's simulation tools are interactive digital tools and online systems that are designed to increase public understanding of policy issues. They have been designed for local governments to educate residents on complex budgetary and policy trade-offs, foster consensus, and collect data-driven community feedback. These simulations provide real-time, visual impacts of decisions, enabling officials to align public priorities with fiscal realities by allowing users to manipulate variables—such as increasing tax revenue or cutting specific services.

Several examples of awareness platforms include:

- **CitizenLab:** Focuses on local governments and support idea collection, surveys, participatory budgeting, and policy co-creation with strong analytics.

- **Bang the Table:** This is developed by EngagementHQ they are used for formal consultations, project communication, and community building; emphasizes data-driven reporting.

- **Pol.is:** This is used for large-scale AI-assisted awareness and surveys; maps opinions to reveal consensus rather than polarization.

- **CONSUL:** An open-source platform for proposing/voting policies used globally for participatory policymaking.

- **Policy Simulations**: This is developed by socialsimulations.org focuses on interactive, participatory activities using social simulation methods. Participants use real-life roles, real-world data, and cultural contexts to explore actual policy issues (e.g., flood disaster risk management). It is geared toward collaborative goal setting and strategy testing for real-world applicability.

Polco Simulation Tools: Community-oriented interactive tools for public engagement in budgeting, housing, land use, and policymaking. Includes budget simulators where residents experiment with allocation decisions to understand trade-offs.

Other engagement platforms include WeSolve, Civocracy, OpenGov, Fujitsu Policy Twin, Nesta Policy Atlas. These engagement platforms are all supportive of ongoing conversations, challenges, ideation, and integration with government processes.

Specialized Domains

These are computational, agent-based, or virtual tools for modeling domain-specific complex systems such as healthcare and urban planning etc. They test the impacts, risks, and benefits of specific policy outcomes and resource allocations before implementation. These tools also function as experimental platforms, enabling policymakers to evaluate potential outcomes and identify synergies and conflicts that can arise between policies. There are several examples of specialized domain simulators, which are outlined accordingly:

- **C-ROADS (CLIMATE INTERACTIVE):** This is a virtual policy simulator that addresses climate change policies, including national greenhouse gas reduction strategies and global warming. Policymakers have widely used it to contextualize landmark achievements such as the 2015 Paris Climate Change Agreement.

- **En-ROADS ():** This is a specialized domain simulator that is inextricably linked to C-ROADS for broader energy/climate policy.
- **Energy Policy Simulator (policysimulator.org)**: A web-based tool used to model energy, climate, and related policies, as well as to determine sector-specific outcomes.
- **RAND Concurrent Policies Simulation Tool**: This is a tool that effectively explores how statistical models handle overlapping state-level policies, as well as evaluates under concurrent implementations.
- **City Digital Twins** (e.g., Singapore). This is an AI-enhanced tool that is used to address such disasters as emergencies

Table 3 from Grok shows policy platforms grouped by their underlying AI-driven technology.

Next, as a case study, we present a "full-stack" architecture for an AI-driven policy simulator.

1. **AI-driven Policy Simulator architecture**

Our As discussed, our goal is to implement a holistic AI-driven policy simulator.

Designing an AI-driven policy simulation platform involves creating a modular, scalable, transparent, and participatory system that leverages generative AI, agent-based modeling, and simulation techniques to help policymakers, researchers, and citizens explore complex policy impacts before real-world implementation. The goal is to move beyond static models toward dynamic, human-like behavioral simulations that reveal emergent effects, unintended consequences, and equity implications.

Policy Simulation architecture overview:

Figure 2 captures a "full stack" architecture for the AI-driven policy simulator.

At the top level we have the Stakeholder User Interface (UI). Next the Policy Intervention layer defining the policy being simulated and the various outcomes. This is followed by the simulator experimentation and engagement layer.

Below we have the data and knowledge layer. Then the simulation core layer comprising the Agentic AI and multi-agent algorithms. Lastly, we have an orchestration and execution engine that runs the simulation.

Challenges, concerns and risks associated with policy simulation divide into technical challenges, ethical challenges, societal and policy risks, and governance/oversight issues:

Technical Challenges

- Validation and reliability gaps: Agents are benchmarked against human self-consistency (which itself varies), but performance drops unseen tasks, non-textual stimuli, or highly dynamic environments. Interview-based agents outperform simpler baselines, yet they still underperform for certain demographics (e.g., conservatives vs. liberals on some GSS items) and cannot fully capture long-horizon or black-swan events.
- Scale vs. expressiveness trade-off: Full LLM agents are computationally prohibitive at population scale (millions of agents). MIT Media Lab's work on "LLM archetypes" (heuristic + adaptive hybrids) enables 8.4 million-agent COVID-19 simulations for New York City, but realism suffers when expressiveness is sacrificed for feasibility.
- Non-determinism and emergence: Small prompt variations or LLM stochasticity can produce divergent policy outcomes. Sensitivity analysis (e.g., Monte Carlo runs) is essential but expensive; black-box reasoning makes causal attribution difficult.

Ethical Concerns

- **Privacy and re-identification**: Agents are built from sensitive, hard-to-anonymize interview transcripts containing personal histories, demographics, and attitudes. Even pseudonymized data risks leakage or future inference by more powerful models. Stanford's team explicitly warns of re-identification and requires 25-year withdrawal rights.
- **Consent and data stewardship**: Participants consent knowing full anonymity is "challenging," but downstream policy use raises questions of ongoing control. Misuse could generate simulated "statements" falsely attributed to real individuals.
- **Value embedding and bias amplification**: Agents inherit training-data or interview biases. While interview-based designs reduce stereotyping compared to demographic personas, residual disparities remain; policy simulations could inadvertently encode designer values (e.g., efficiency over equity).

- **Responsibility gaps**: If a simulation recommends a flawed policy, who is liable—model developers, data providers, or policymakers who defer to "the agents"?

Societal and Policy Risks

- **Over-reliance and deskilling**: Policymakers may treat agents as oracles rather than exploratory tools, eroding human judgment. Stanford cautions against "harms related to over-reliance on agents."
- **Reputation and misuse harms**: Malicious actors could weaponize agents for disinformation, synthetic public opinion, or targeted manipulation—e.g., simulating "voter backlash" to kill regulations. Worst-case scenarios include defamatory fabricated responses.
- **Unintended consequences and inequity**: Simulations may mask disparate impacts on underrepresented groups or amplify second-order effects (e.g., a tax policy that entrenches inequality). Scalability limits mean many real-world complexities remain unmodeled.
- **Security vulnerabilities**: Agentic systems inherit prompt-injection, model-poisoning, or autonomy risks; a compromised simulation could output manipulated policy advice.

Governance and Oversight Issues

- **Transparency and auditability**: Multi-step agent reasoning is opaque; mandatory explainability, uncertainty quantification (e.g., confidence intervals), and independent audits are needed but rare.
- **Access and commercialization controls**: Stanford's two-pronged system (open aggregated benchmarks + restricted individual agents under IRB review and non-commercial agreements) is a model, but scaling this across proprietary tools remains unresolved.
- **Regulatory fragmentation**: No unified global standards exist for simulation use in policymaking; EU AI Act high-risk classifications may apply, but enforcement lags.
- **Commercial influence**: Private labs or consultants could design simulations favoring certain outcomes.

Table 4 reviews AI-driven Policy Simulation challenges and risks.

Conclusion

AI-driven policy simulation platforms are powerful experimental models designed to enhance engagement and decision-making during policy implementation. The authors of this paper acknowledge concerns with using AI to simulate policies (e.g., hallucinations and data contamination). However, these challenges can be mitigated through continuous monitoring, diverse validation, and precautionary governance by deploying a human-in-the-loop approach to actively prevent the amplification of inequalities in policies.

Hence, it is evident that AI-driven policy simulation platforms are an innovative approach in the context of the ongoing 4th Industrial Revolution. The authors of this paper therefore strongly recommend that these AI tools can be effectively utilized to address major policy issues facing national governments in the 2020s. In summary:

- **Technical & Methodological:** The development of the AI-driven policy simulation platform uses the latest GenAI and Agentic AI, as well as multi-agent modeling technology. This posed a challenge to the framework implemented in this research project, as the most recent AI models lack real-world training data.
- **Data & Validation:** As this research project used synthetic model data, there was concern that the training processes for the AI-driven policy simulation platform might be contaminated. The authors of this paper were therefore keen to address this issue, which often occurs unintentionally during AI training.

- **Operational & Implementation:** During this research project, the authors engaged with interpretability by providing natural-language explanations of emergent outcomes. This was challenging as continuous auditing of agent demographics requires diverse training prompts and equity metrics.
- **Ethical, Societal, & Governance:** the authors of this paper considered the ethical challenges associated with deploying the human-in-the-loop approach. One such challenge was that the authors were concerned about the inclusion of human bias and AI's susceptibility to hallucinations and misinterpretations.

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